

Describing Distributions

In AP Statistics, to describe a distribution, we use the acronym SOCS:

S:

O:

C:

S:

After we graph a quantitative variable, describing what we see helps identify patterns, and answer questions we might have about the data.

****Note:** We do NOT use SOCS to describe categorical variables. Why?

Categories can be placed in any order on the x-axis, so shape wouldn't make sense. Same thing for center and spread, we cannot find the average zip code (for example)

Shape

Once the distribution is graphed, the first thing we identify is the shape of the graph.

Unimodal	Bimodal	Uniform
Symmetric	Left Skewed	Right Skewed

You might only use one word to describe the shape or two words might be appropriate:

Draw a symmetric, bimodal distribution.	Draw a unimodal, right skewed distribution.

Outliers

An **outlier** is an individual piece of data that falls outside the overall pattern of the distribution.

When an outlier occurs, we must find out *why* it occurs. Many times it occurs because of a mistake. It may have been written down incorrectly or input into a computer or calculator incorrectly. That is why we always PLOT OUR DATA first because our eye will usually tell us when an outlier exists. Outliers can be eliminated from the data if there is a good reason.

It isn't that easy to determine whether or not to eliminate an outlier and different statisticians may handle the situation differently. But with outliers, there are several rules:

- 1) Plot the data
- 2) Determine whether there is an outlier. If so, be sure that the data is accurate.
- 3) Determine if there is a valid reason that the outlier can be removed. If there isn't, don't remove it just identify it.

Center

There are three measures of center in statistics: _____, _____, and _____.

Let's gather some data to analyze the measures of center: How many siblings (step and half included, because they are!) do you have? Copy down the class data below.

I'm sure you know how to find the "average" (technically, average is a generic term for a measure of center. When you say average, you usually are referring to the **mean**. We do not use the term "average" in AP Statistics), but here is the formula and notation:

	_____ : mean of a sample _____ : each individual observation _____ : number of observations _____ : capital sigma = "add up"
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Mean number of siblings in our class: _____

The reason we do not always use the mean to describe the center has to do with the inclusion of outliers in a data set.

For instance, suppose a student took 7 exams and had these scores: 90, 92, 94, 98, 86, 88, 0.

- There is very clearly an outlier: the zero score. There might be legitimate reasons to eliminate the outlier. Why?
- If we cannot eliminate the outlier, we calculate the mean to be: _____
- If we did eliminate the outlier, we calculate the mean to be: _____
- What is the outlier doing to the mean? _____

The mean is called **non-resistant**. This means that the mean is strongly influenced by extreme values. If you do have extreme values in a data set (these extreme values may or may not be outliers), the mean is not reliable as a measure of center.

Luckily, we have another measure of center that is **resistant**: meaning that if there are extreme values in a data set, this measure of center will not be as affected by it.

The **median** is found by ordering the data and then finding the middle value in that list.

For example, order the number of siblings from lowest to highest:

Mean: _____ Median: _____ How do these compare? _____

Let's take a look back at our exam score example: 0, 86, 88, 90, 92, 94, 98

Mean: _____ Median: _____ How do these compare? _____

****Note:** This isn't saying that the median is always a better measure of center, but whenever there are extreme values or outliers, the median is the better measure of center.

****Note:** The median does not have a formula or special math symbol. (Use Med for short)

The last measure of center is perhaps the most useless: the mode. It finds the most occurring value in a data set. The idea of having this as a measure of center comes from having a symmetric, unimodal distribution, where the most occurring value happens in the middle. However, as we have seen, there are many shapes to different distributions and the most occurring doesn't always occur in the middle.

78, 79, 80, 80, 85, 91, 99	77, 77, 78, 79, 85, 100	73, 73, 88, 89, 90, 93, 98
Mean: _____ Median: _____ Mode: _____ Shape: _____	Mean: _____ Median: _____ Mode: _____ Shape: _____	Mean: _____ Median: _____ Mode: _____ Shape: _____

To summarize our mini exploration above:

<i>Unimodal, Symmetric</i>	<i>Left Skewed</i>	<i>Right Skewed</i>

Spread

There are three common measures of spread in statistics: _____, _____, and _____.

The **range** is the difference: _____ - _____ = _____. The range is a number, not an interval!

The **standard deviation** is the average deviation of an observation from the mean of the data set. To understand where the formula comes from, let's walk through an example where we find the standard deviation by hand (which we will rarely have to do):

Example: A group of elementary school children was asked how many pets they have. Here are their responses, arranged from highest to lowest:

1 3 4 4 4 5 7 8 9

Calculate the mean: _____

Observations	1	3	4	4	4	5	7	8	9	
Deviations										
Square Deviations										

Now we will compute the average squared deviation – sort of. Instead of dividing by the total number of observations, we will divide by $n-1$ (the reason for this involves some calculus, so just trust me on this one).

This value, _____, is called the **variance** (S_x^2). This is the average squared deviation of the number of pets this group of elementary children have.

Because this is the average squared deviation, our units are in “squared pets”, and that is pretty useless to us.

We have to take the square root to get back to the correct units – pets.

Variance = _____ $\rightarrow$ _____ $\rightarrow$ Standard Deviation (S_x) = _____

The variance formula is:

	_____ : symbol for variance
	_____ : mean of a sample
	_____ : each individual observation
	_____ : number of observations
	_____ : capital sigma = “add up”
	_____ : symbol for standard deviation

The standard deviation formula is very similar, and you can see the difference is just undoing the “square” by square rooting both sides of the formula:

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Interpreting the standard deviation in context:

- The number of pets typically varies from the mean by _____.
- It measures the typical distance of the values in a distribution from the mean.

Standard deviation should only be used as your measure of spread when the mean is your chosen measure of center.

The standard deviation has the following additional properties

- The standard deviation is always _____.
- The standard deviation is 0 when _____.
- The standard deviation has the same units of measure as the original variable measured.
- The standard deviation is _____, meaning that a few outliers will make it _____.
- The greater the standard deviation, the greater the variation of the distribution.

The last measure of spread is **IQR**. IQR stands for *Inter-Quartile Range* and uses percentiles to describe the spread of the distribution.

Recall: **Percentiles** measure the percent of the observations that fall below a value.

If the median is the middle of a data set (50% below and 50% above), what percentile is it? _____

Here are some other percentiles:

- 0th percentile: _____ - lowest value in a data set.
- 25th percentile: _____ or _____ - 25% of the data is below this value.
- 50th percentile: _____ - middle value in a data set.
- 75th percentile: _____ or _____ - 75% of the data is below this value.
- 100th percentile: _____ - highest value in a data set.

These 5 numbers make up what is called the Five Number Summary (clever, right?). To calculate these values by hand, place the observations in order and find the median. Then, find the middle value to the left and right of your median to identify your quartiles.

Example – Here is the data from a previous AP statistics class. They were asked how many hours of sleep they got before the first day of school.

2 4.5 5 5 6 6 6 6.5 7 7 7 7 7 7 7 7.5 8 8 8 8 8 8.5

Min : _____ Q1: _____ Median: _____ Q3: _____ Max: _____

The IQR is the range of the inter-quartiles: _____ Just like range, this value is a SINGLE value. It is not defined as an interval.

What is the IQR of the amount of hours of sleep that group of AP Stat students got? _____.

Boxplots

The five number summary can also be used to create another graph: the **boxplot**.

Example: Using the five number summary above, create a boxplot of the amount of hours of sleep the AP stat students got.

Using the five number summary and the IQR, we also have a numerical way of determining if a point is an outlier. For simplicity's sake, we call this the **outlier test**.

Steps:

1. Find the 5-number summary
2. Find the IQR
3. Compute $Q1 - (1.5 * IQR)$. Any data below that number is an outlier.
4. Compute $Q3 + (1.5 * IQR)$. Any data above that number is an outlier.

Using the sleep hours in the example above, is there an outlier? Show your work.

When we have outliers in our data set, and we cannot throw them out, a **modified boxplot** helps us create a boxplot that will identify the outliers. Using the example above, once we have determined that we have an outlier, we mark it's location with a star (or dot). Then, instead of the whisker extending to the outlier, it extends to the next observation that is within the acceptable range.

Example – create a modified boxplot for the sleep example above.

Shape of Boxplot

Distribution	Boxplot	Key Features
		Symmetric Uniform mean $\approx$ median
		Symmetric Bimodal mean $\approx$ median
		Symmetric Unimodal Bell-Shaped mean $\approx$ median
		Unimodal Left Skewed mean $<$ median
		Unimodal Right Skewed mean $>$ median